

Random Forest

Εκπαιδεύουμε πολλά variations του μοντέλου με διαφορετικές υπερπαραμέτρους και βρίσκουμε το μοντέλο με τις καλύτερες υπερπαραμέτρους. Ύστερα, το εφαρμόζουμε πάνω στο test gold για να κάνουμε προβλέψεις και τις αποθηκεύουμε για να γίνουν evaluate μαζί με όλες τις άλλες

```
from pyspark.sql import SparkSession
import pyspark.sql.functions as F
from pyspark.sql.types import DoubleType
from pyspark.ml.feature import VectorAssembler
from pyspark.ml.classification import RandomForestClassifier
from pyspark.ml.tuning import ParamGridBuilder, CrossValidator
from pyspark.ml.evaluation import BinaryClassificationEvaluator

print("1. Εκκίνηση SparkSession...")
spark = SparkSession.builder \
    .appName("RF_SMOTE_Corrected_GridSearch") \
    .master("local[*]") \
    .config("spark.driver.memory", "8g") \
    .getOrCreate()

print("2. Φόρτωση και Feature Augmentation...")
train_gold = spark.read.parquet("../data/train_gold_with_cluster.parquet")
test_gold = spark.read.parquet("../data/test_gold_with_cluster.parquet")

train_gold = train_gold.withColumn("stroke",
train_gold["stroke"].cast(DoubleType()))
test_gold = test_gold.withColumn("stroke",
test_gold["stroke"].cast(DoubleType()))
train_gold = train_gold.withColumn("cluster",
F.col("cluster").cast(DoubleType()))
test_gold = test_gold.withColumn("cluster", F.col("cluster").cast(DoubleType()))

assembler = VectorAssembler(inputCols=["features", "cluster"],
outputCol="augmented_features")
train_augmented =
assembler.transform(train_gold).drop("features").withColumnRenamed("augmented_features",
"features")
test_augmented =
assembler.transform(test_gold).drop("features").withColumnRenamed("augmented_features",
"features")

# Κρατάμε ολόκληρο το SMOTE dataset (χωρίς undersampling)
train_augmented.cache()

print("3. Ορισμός Πλέγματος Παραμέτρων (Grid Search)...")
rf_base = RandomForestClassifier(featuresCol="features", labelCol="stroke",
seed=22390225)
```

```

# ΔΙΚΑΙΟΛΟΓΗΣΗ (Η ΔΙΟΡΘΩΣΗ):
# Κατεβάζουμε το maxDepth στα 4, 6 και 8. Αυτό εμποδίζει το δέντρο να
# "αποστηθείσει" τα συνθετικά δεδομένα του SMOTE, κρατώντας το Recall ψηλά
# για τα πραγματικά, άγνωστα δεδομένα του Test Set.
paramGrid = (ParamGridBuilder()
              .addGrid(rf_base.maxDepth, [4, 6, 8])
              .addGrid(rf_base.numTrees, [100, 200])
              .build())

# Χρήση PR-AUC για την αξιολόγηση
evaluator = BinaryClassificationEvaluator(labelCol="stroke",
rawPredictionCol="rawPrediction", metricName="areaUnderPR")

cv = CrossValidator(estimator=rf_base,
                    estimatorParamMaps=paramGrid,
                    evaluator=evaluator,
                    numFolds=3,
                    seed=22390225)

print("4. Εκτέλεση Cross-Validation στο ΠΛΗΡΕΣ SMOTE Dataset...")
cv_model = cv.fit(train_augmented)
best_rf = cv_model.bestModel

print("\n=====")
print("[ΔΙΟΡΘΩΜΕΝΗ ΕΥΡΕΣΗ SMOTE] Οι βέλτιστες παράμετροι βρέθηκαν:")
print(f" -> maxDepth: {best_rf._java_obj.getMaxDepth()}")
print(f" -> numTrees: {best_rf._java_obj.getNumTrees()}")
print("=====")

print("5. Παραγωγή και αποθήκευση προβλέψεων στο Test Set...")
rf_preds = best_rf.transform(test_augmented)

# Αποθήκευση ως το κεντρικό rf_predictions
rf_preds.select("stroke", "prediction",
               "probability").write.mode("overwrite").parquet("../data/
rf_predictions.parquet")

spark.stop()
print("Ολοκληρώθηκε! Έχεις πλέον ένα RF που αντιστέκεται στο overfitting του
SMOTE.")

1. Εκκίνηση SparkSession...

WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/06/09 22:59:42 WARN Utils: Your hostname, cachyos-x8664, resolves to a
loopback address: 127.0.1.1; using 192.168.1.5 instead (on interface enp4s0)
26/06/09 22:59:42 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another
address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties

```

Setting default log level to "WARN".

To adjust logging level use `sc.setLogLevel(newLevel)`. For SparkR, use `setLogLevel(newLevel)`.

26/06/09 22:59:43 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

2. Φόρτωση και Feature Augmentation...

3. Ορισμός Πλέγματος Παραμέτρων (Grid Search)...

4. Εκτέλεση Cross-Validation στο ΠΛΗΡΕΣ SMOTE Dataset...

26/06/09 22:59:53 WARN DAGScheduler: Broadcasting large task binary with size 1506.8 KiB

26/06/09 22:59:53 WARN DAGScheduler: Broadcasting large task binary with size 1326.2 KiB

26/06/09 22:59:54 WARN DAGScheduler: Broadcasting large task binary with size 1252.1 KiB

26/06/09 22:59:54 WARN DAGScheduler: Broadcasting large task binary with size 1881.4 KiB

26/06/09 22:59:55 WARN DAGScheduler: Broadcasting large task binary with size 1351.3 KiB

26/06/09 22:59:56 WARN DAGScheduler: Broadcasting large task binary with size 1506.8 KiB

26/06/09 22:59:56 WARN DAGScheduler: Broadcasting large task binary with size 2.3 MiB

26/06/09 22:59:56 WARN DAGScheduler: Broadcasting large task binary with size 3.6 MiB

26/06/09 22:59:57 WARN DAGScheduler: Broadcasting large task binary with size 2.5 MiB

26/06/09 23:00:01 WARN DAGScheduler: Broadcasting large task binary with size 1507.3 KiB

26/06/09 23:00:01 WARN DAGScheduler: Broadcasting large task binary with size 1353.3 KiB

26/06/09 23:00:02 WARN DAGScheduler: Broadcasting large task binary with size 1255.0 KiB

26/06/09 23:00:02 WARN DAGScheduler: Broadcasting large task binary with size 1881.6 KiB

26/06/09 23:00:03 WARN DAGScheduler: Broadcasting large task binary with size 1352.5 KiB

26/06/09 23:00:04 WARN DAGScheduler: Broadcasting large task binary with size 1507.3 KiB

26/06/09 23:00:04 WARN DAGScheduler: Broadcasting large task binary with size 2.4 MiB

26/06/09 23:00:04 WARN DAGScheduler: Broadcasting large task binary with size 3.6 MiB

26/06/09 23:00:05 WARN DAGScheduler: Broadcasting large task binary with size 2.6 MiB

26/06/09 23:00:09 WARN DAGScheduler: Broadcasting large task binary with size 1496.7 KiB

26/06/09 23:00:09 WARN DAGScheduler: Broadcasting large task binary with size 1314.3 KiB

26/06/09 23:00:10 WARN DAGScheduler: Broadcasting large task binary with size 1268.2 KiB
26/06/09 23:00:10 WARN DAGScheduler: Broadcasting large task binary with size 1900.3 KiB
26/06/09 23:00:11 WARN DAGScheduler: Broadcasting large task binary with size 1362.1 KiB
26/06/09 23:00:11 WARN DAGScheduler: Broadcasting large task binary with size 1496.7 KiB
26/06/09 23:00:12 WARN DAGScheduler: Broadcasting large task binary with size 2.3 MiB
26/06/09 23:00:12 WARN DAGScheduler: Broadcasting large task binary with size 3.5 MiB
26/06/09 23:00:13 WARN DAGScheduler: Broadcasting large task binary with size 2.5 MiB
26/06/09 23:00:14 WARN DAGScheduler: Broadcasting large task binary with size 1563.2 KiB
26/06/09 23:00:14 WARN DAGScheduler: Broadcasting large task binary with size 2.5 MiB
26/06/09 23:00:14 WARN DAGScheduler: Broadcasting large task binary with size 3.9 MiB

```
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[ΔΙΟΡΘΩΜΕΝΗ ΕΥΡΕΣΗ SMOTE] Οι βέλτιστες παράμετροι βρέθηκαν:
-> maxDepth: 8
-> numTrees: 200
=====
```

5. Παραγωγή και αποθήκευση προβλέψεων στο Test Set...

26/06/09 23:00:15 WARN DAGScheduler: Broadcasting large task binary with size 2.9 MiB

Ολοκληρώθηκε! Έχεις πλέον ένα RF που αντιστέκεται στο overfitting του SMOTE.